Arduino Temperature Monitoring & Alert System Report

Project Title: Temperature Monitoring System with DS18B20 Sensor, 16x2 LCD Display, and LED Alert

Simulation Link: Wokwi Project Link

Threshold Temperature: 25.0 °C

1. Introduction & Objective

This project implements an embedded temperature monitoring and warning system using an Arduino Uno microcontroller. The system continuously reads ambient temperature via a Dallas DS18B20 digital temperature sensor, displays real-time temperature data and status messages on a 16x2 Liquid Crystal Display (LCD), and triggers an LED warning indicator when the temperature exceeds a predefined threshold of 25.0 °C.

2. Components Required

Component Name	Quantity	Purpose / Description
Arduino Uno	1	Main microcontroller board running the control logic.

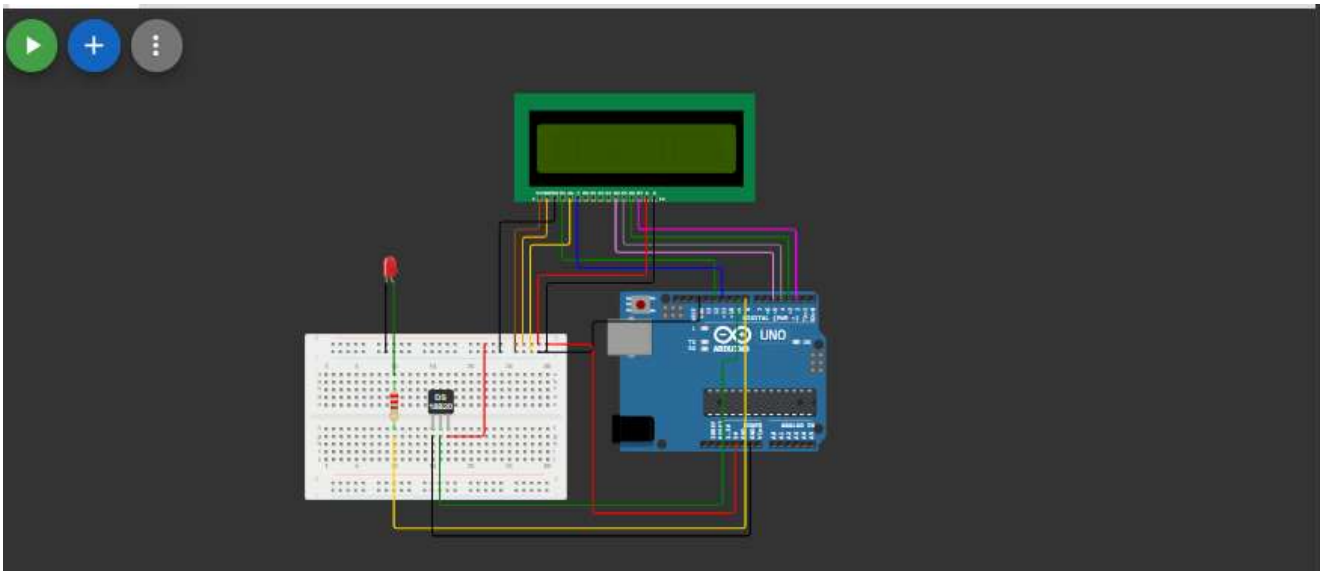
Component Name	Quantity	Purpose / Description
DS18B20 Temperature Sensor	1	Digital 1-wire temperature sensor for accurate ambient measurement.
16x2 LCD Display	1	Visual interface to show current temperature and alert statuses.
Resistor (220Ω current-limiting)	1	220Ω resistor for the LED.
LED	1	Visual alert indicator that turns ON when temperature exceeds 25 °C.
Breadboard & Jumper Wires	1 & As required	For circuit prototyping and interconnections.

3. Circuit Connections & Wiring Guide

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  DS18B20 Sensor: Powered via 5V and GND. The Data pin is connected to Digital Pin 9 on the Arduino Uno..
- **16x2 LCD Display (Parallel Mode):**
  - RS pin -> Digital Pin 12
  - Enable (E) pin -> Digital Pin 11
  - D4, D5, D6, D7 -> Digital Pins 5, 4, 3, 2

    - VSS to GND, VDD to 5V, V0 to potentiometer/contrast, RW to GND, Backlight anode (A) to 5V through current limiting, Backlight cathode (K) to GND.
- **LED Alert Circuit:** Anode connected via a 220Ω current-limiting resistor to Digital Pin 8, and Cathode connected to GND.



4. Arduino Source Code

```
#include <LiquidCrystal.h>
#include <OneWire.h>
#include <DallasTemperature.h>

// LCD: RS, E, D4, D5, D6, D7
LiquidCrystal lcd(12, 11, 5, 4, 3, 2);

const int sensorPin = 9;
const int ledPin = 8;
const float threshold = 25.0;

OneWire oneWire(sensorPin);
```

```
DallasTemperature sensors(&oneWire);

void setup() {
 pinMode(ledPin, OUTPUT);
 digitalWrite(ledPin, LOW);

 lcd.begin(16, 2);
 lcd.print("Temperature");
 lcd.setCursor(0, 1);
 lcd.print("Monitoring...");

 sensors.begin();
 Serial.begin(9600);

 delay(1500);
 lcd.clear();
}

void loop() {
 sensors.requestTemperatures();

 float temperatureC = sensors.getTempCByIndex(0);

 if (temperatureC == DEVICE_DISCONNECTED_C) {
  digitalWrite(ledPin, LOW);
  lcd.setCursor(0, 0);
  lcd.print("Sensor Error   ");
  lcd.setCursor(0, 1);
  lcd.print("Check DS18B20  ");
  Serial.println("DS18B20 not detected");
  delay(1000);
  return;
 }

 bool highTemperature = temperatureC > threshold;

 digitalWrite(ledPin, highTemperature ? HIGH : LOW);
```

```
  lcd.setCursor(0, 0);
  lcd.print("Temp: ");
  lcd.print(temperatureC, 1);
  lcd.write((uint8_t)223);
  lcd.print("C   ");

  lcd.setCursor(0, 1);
  if (highTemperature) {
    lcd.print("ALERT: HIGH TEMP");
  } else {
    lcd.print("Status: NORMAL  ");
  }

  Serial.print("Temperature: ");
  Serial.print(temperatureC, 1);
  Serial.println(" C");

  delay(1000);
}
```

5. Step-by-Step Working & Explanation

1. ---

   Initialization: Upon startup, the LCD initializes displaying "Temperature Monitoring...", while the DS18B20 sensor and serial communication at 9600 baud are configured.
2. **Temperature Request:** In every loop iteration, the Arduino commands the DS18B20 sensor to initiate a temperature conversion via the 1-wire protocol.
3. **Error Handling:** The system checks if the sensor is connected properly (`DEVICE_DISCONNECTED_C`). If disconnected, it displays an error message on the LCD and Serial Monitor.
4. **Threshold Evaluation:** The read temperature is compared against the threshold limit .

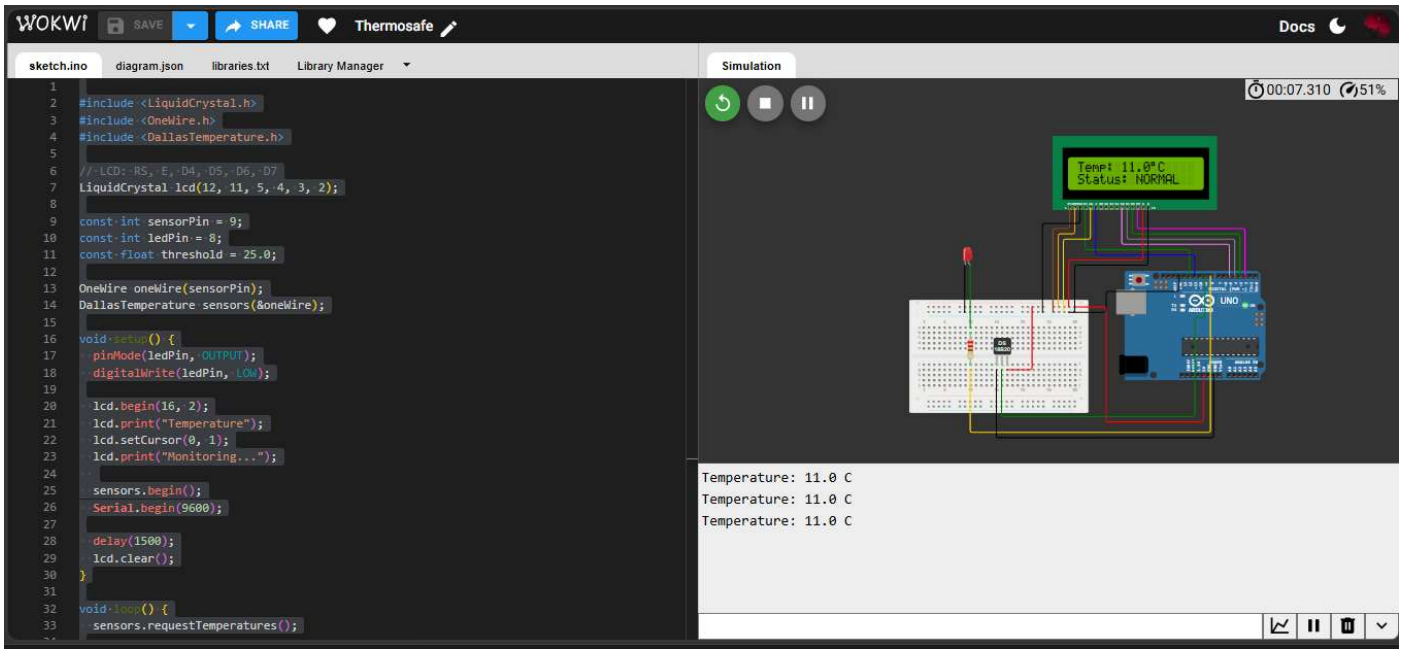
5. **Actuation & Display:**
    - If temperature > 25.0 °C, the LED turns ON and the LCD displays an alert message ("ALERT: HIGH TEMP").
    - If temperature ≤ 25.0 °C, the LED remains OFF and the LCD shows "Status: NORMAL".

6. Simulation Output & Verification

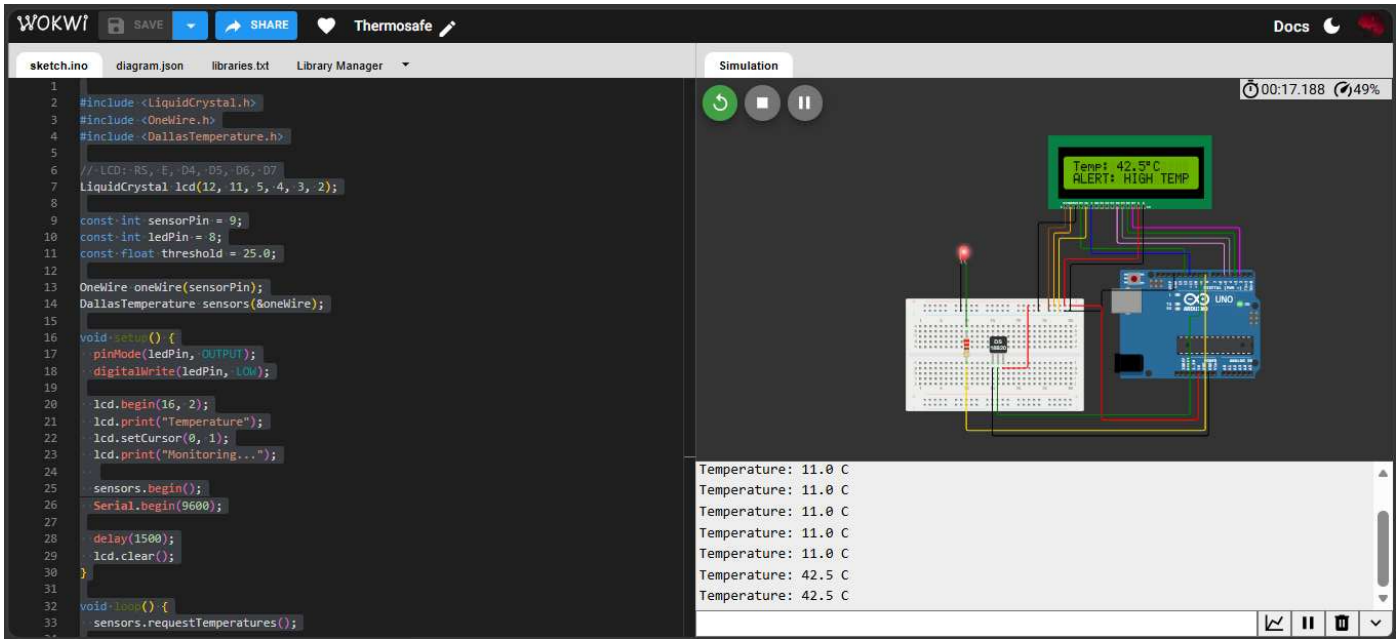
The project was successfully built and tested in Wokwi. Below are summaries of the simulation behavior:

- **Normal State:** When simulated temperature is below 25.0 °C (e.g., 20.3 °C), the LCD displays the temperature with normal status, and the alert LED remains off.
- **Alert State:** When the slider is adjusted above 25.0 °C (e.g., 42.5 °C), the LED lights up immediately and the LCD updates to display "ALERT: HIGH TEMP".
- The following are the screenshots of the simulation run:

1) CASE 1: When the temperature is below 25 degrees:



21 CASE 2: When the temperature is above 25 degrees:



- **It is observed that the LED glows when the temperature exceeds 25 degrees , the warning red LED glows and the text is displayed in the LCD screen.**